

Total No. of Printed Pages: 01

**SUBJECT CODE NO: - RNP-8013**  
**FACULTY OF SCIENCE AND TECHNOLOGY**  
**F.E. (Group A & B) Mech/ Civil / CSE / AI & ML / EEE (NEP) Sem - I**  
**Examination June/July 2025**  
**Design Thinking**

[Time: 2 Hours]

[Max. Marks: 30]

“Please check whether you have got the right question paper.”

**N. B**

1. Part A: Question No 1 is compulsory.
2. Attempt any four questions from remaining Part B: Q.2 to Q.7.

**Part A**

**Q1 Attempt any Five**

**10**

1. **What is the first step in the Design Thinking process?**
  - a) Prototyping
  - b) defining the problem
  - c) Empathizing with the user
  - d) Testing the solution
2. **Design Thinking is most effective when applied to problems that are:**
  - a) Simple and straightforward
  - b) Ill-defined or complex
  - c) Purely technical
  - d) Well-documented with clear solutions
3. **Empathy maps are divided into which categories?**
  - a) Technical, Operational, Financial, Emotional
  - b) Thinking, Doing, Observing, Reacting
  - c) Says, Thinks, Feels, Does
  - d) Needs, Wants, Actions, Reactions
4. **The process of repeatedly asking "Why?" to uncover the root cause of a problem is called:**
  - a) Brainstorming
  - b) Root cause analysis
  - c) Empathy mapping
  - d) 5 Whys technique
5. **Which of the following is NOT an ideation technique?**
  - a) Brainstorming
  - b) Mind mapping
  - c) SWOT analysis
  - d) BrainwritingState
6. Design Thinking is a \_\_\_\_\_-centered approach to problem-solving.
7. \_\_\_\_\_ is a group activity where participants generate ideas without judgment.

**Part B**

**Solve any four from the following**

- |                                                       |           |
|-------------------------------------------------------|-----------|
| <b>Q2</b> Explain Design thinking process.            | <b>05</b> |
| <b>Q3</b> What are the challenges in design thinking? | <b>05</b> |
| <b>Q4</b> Explain Empathy mapping.                    | <b>05</b> |
| <b>Q5</b> Explain Define phase.                       | <b>05</b> |
| <b>Q6</b> Explain Mind mapping.                       | <b>05</b> |
| <b>Q7</b> Write a note on benefits of prototyping.    | <b>05</b> |